

# Lightning Games

One-page repo summary generated from local project evidence on 2026-04-13. Scope is limited to files present in this repository.

Electron 28.0.0

59 game files

Tray + hotkey launcher

localStorage persistence

Bun / npm scripts

## WHAT IT IS

- Lightning Games is an Electron desktop app that lives in the system tray.
- It opens a frameless launcher window with a neon styled interface.
- The launcher is designed for quick access to a large local game library.
- The repo currently ships 59 JavaScript game files under the games folder.
- Games are loaded into a single HTML shell that also hosts the launcher UI.
- The app runs offline from local assets packaged with Electron.
- It uses a secure preload bridge instead of exposing Node APIs to the renderer.
- The main process manages the tray icon, global hotkey, window visibility, and quit flow.
- The renderer manages game registration, launch, pause, resume, and score updates.
- Player progress is stored in browser localStorage rather than a server database.
- The interface includes search, categories, favorites, recent activity, and stats views.
- The experience is heavily themed with particle effects, glow treatments, and multiple themes.

## WHO IT IS FOR

Primary user: a Windows desktop player who wants instant, low-friction access to short local games from the tray.

Likely persona: someone who prefers offline play, quick breaks, visible progress tracking, and a playful premium UI without accounts or setup overhead.

## HOW TO RUN

- 1. From the repo root, install dependencies with `bun install` or `npm install`.
- 2. Start the app with `bun start` or `npm start`.
- 3. Use `bun run dev` or `npm run dev` if you want DevTools.
- 4. Build a distributable with `bun run dist` or `npm run dist`.

## HOW IT WORKS

- Main process: main.js configures GPU and privacy-related Chromium flags, creates the BrowserWindow, manages the tray menu, registers Ctrl+Alt+G, and handles close/quit IPC.
- Preload bridge: preload.js exposes only closeWindow, quitApp, and window show/hide event hooks through contextBridge.
- Renderer shell: index.html defines the launcher, stats, settings, data modals, game canvas, and script includes for renderer modules plus all game scripts.
- Game lifecycle: renderer/gameManager.js registers game classes, instantiates the active game, runs the animation loop, pauses/resumes, updates scores, and checks achievements.
- UI layer: renderer/launcher.js renders game cards from GAME\_CARDS\_CONFIG, wires search, category tabs, favorites, stats, settings, export/import, screenshot, and tutorial flows.
- Media layers: renderer/soundManager.js synthesizes music and SFX with the Web Audio API; renderer/particles.js draws the animated background canvas.
- Data flow: user input in launcher -> gameManager launches/updates games -> score and stats mutate in memory -> localStorage persists highscores, settings, achievements, favorites, playtime, and last-played data.
- Remote services, cloud sync, and external APIs: Not found in repo.

## WHAT IT DOES

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| <ul style="list-style-type: none"><li>• Electron desktop runtime</li><li>• System tray presence</li><li>• Tray click toggles window</li><li>• Global hotkey Ctrl+Alt+G</li><li>• Single-instance lock</li><li>• Frameless fixed-size window</li><li>• Always-on-top launcher window</li><li>• Hide instead of full close</li><li>• Auto-hide on blur</li><li>• Startup fade-in flow</li><li>• Secure preload bridge</li><li>• Context isolation enabled</li><li>• Node integration disabled</li><li>• Web security enabled</li><li>• 59 local game scripts included</li><li>• Arcade game category</li><li>• Puzzle game category</li><li>• Classic game category</li><li>• Favorites category tab</li></ul> | <ul style="list-style-type: none"><li>• Searchable game grid</li><li>• Random game button</li><li>• Recently played section</li><li>• Stats dashboard</li><li>• Top games by score</li><li>• Recent activity list</li><li>• Category breakdown</li><li>• Playtime distribution view</li><li>• Milestones panel</li><li>• Achievement list filters</li><li>• Achievement toast container</li><li>• High score tracking</li><li>• Per-game best score display</li><li>• Persistent favorites</li><li>• Persistent achievements</li><li>• Persistent last played timestamps</li><li>• Persistent total play time</li><li>• Persistent consecutive streaks</li><li>• Persistent unique games list</li></ul> | <ul style="list-style-type: none"><li>• Persistent total asteroids stat</li><li>• Volume slider in title bar</li><li>• Procedural SFX engine</li><li>• Per-game music themes</li><li>• Background particle canvas</li><li>• Reduced motion setting</li><li>• Screen shake setting</li><li>• Resolution scale setting</li><li>• Particle density setting</li><li>• Glow intensity setting</li><li>• Animation speed setting</li><li>• Show FPS option</li><li>• Screen flash option</li><li>• SFX volume option</li><li>• Music volume option</li><li>• Mute on blur option</li><li>• Auto-pause option</li><li>• Confirm exit option</li><li>• Show timer option</li></ul> | <ul style="list-style-type: none"><li>• Difficulty selector</li><li>• Compact mode option</li><li>• Show descriptions option</li><li>• Achievement notifications option</li><li>• Card size control</li><li>• Grid layout selector</li><li>• Dark mode toggle</li><li>• Tutorial button</li><li>• JSON data export</li><li>• JSON data import</li><li>• High-score screenshot action</li><li>• Achievement share action</li><li>• Hard reset action</li><li>• Build scripts for Bun and npm</li><li>• Portable Windows build target</li><li>• Linux ApplImage build target</li><li>• Native build UI project included</li><li>• Backend/API services: Not found in repo</li></ul> |
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